

Owen Cornmesser

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EDUCATION

Texas A&M University

Ph.D. Electrical Engineering, Energy and Power Group
Advisor: Dr. Thomas Overbye and Dr. Adam B. Birchfield
Thesis Topic: *Power System Dynamics, Oscillations, and Controls*
GPA: 4.0/4.0

Aug. 2025 – Present
College Station, TX

Colorado School of Mines

M.S. Electrical Engineering, emphasis in Energy and Power Systems
Advisor: Dr. Qiuhua Huang
Thesis: *Adaptive Co-Simulation Tools for Transmission and Distribution Systems with High DER Penetration*
GPA: 4.0/4.0

May 2025
Golden, CO

Colorado School of Mines

B.S. Electrical Engineering, emphasis in Energy and Power Systems
GPA: 3.96/4.0, Summa Cum Laude

May 2024
Golden, CO

PUBLICATIONS AND MANUSCRIPTS

Journal Articles

- [J1] **O. Cornmesser**, S. Chakraborty, and J. Wang, “Control Parameter Sensitivity Study for Inverter-Based Resource Dominated Grids: A Small-Signal Stability Approach and Framework,” *IEEE Transactions on Power Delivery*, under review, Mar. 2026
- [J2] **O. Cornmesser**, S. Chakraborty, and J. Wang, “Influence of Closed-Loop and Open-Loop Power Control on the Interaction Between PLL and Grid Strength,” *IEEE Industry Applications Society*, under review, Jan. 2026

Conference Papers

- [C1] **O. Cornmesser** and Q. Huang, “Power System Dynamic Contingency Analysis with Adaptive Distribution System Modeling,” *to appear in Proceedings of IEEE Power and Energy Society General Meeting*, Jul. 2026.
- [C2] **O. Cornmesser** et al., “Evaluation of a Reduced-Order Model for IBR Fault Response Representation via OEM Blackbox Models,” *to appear in Proceedings of IEEE Power and Energy Society General Meeting*, Jul. 2026.
- [C3] **O. Cornmesser** et al., “Evaluation of a Reduced-Order Model for IBR Fault Response Representation in EMT Fault Studies,” *to appear in Proceedings of the IEEE Transmission and Distribution Conference and Exposition*, May 2026.
- [C4] R. Mahmud, **O. Cornmesser**, et al., “A Data-Driven dq Domain Reduced-Order Model for IBR Fault Models in EMT Studies,” *to appear in Proceedings of the IEEE Transmission and Distribution Conference and Exposition*, May 2026.
- [C5] **O. Cornmesser** and Q. Huang, “Identification of Impact Regions for Adaptive Distribution System Modeling in Transmission-Level Dynamic Contingency Analysis,” *CIGRE Grid of the Future Symposium*, Nov. 2024.

Manuscripts in Preparation

[M1] **O. Cornmesser**, A. B. Birchfield, and T. Overbye, “On Disturbance-Dependent Measurement-Based Modes in Large-Scale Power Systems: A Model-Based Modal Analysis Perspective,” *in preparation for IEEE Transactions on Power Systems*

RESEARCH EXPERIENCE

Texas A&M University

Chancellor’s Fellowship Recipient, Energy and Power Group

Advisors: Dr. Thomas Overbye and Dr. Adam B. Birchfield

Aug. 2025 – Present

College Station, TX

- Develop methods for model linearization of large-scale power systems to identify sources of error in measurement-based modal analysis.
- Analyze how various inverter controls contribute to and/or mitigate system-wide dynamic phenomena using linear and nonlinear analysis.
- Create open-source engineering tools designed for practical implementation by utilities, researchers, and industry users.

National Laboratory of the Rockies (formerly NREL)

Inverter-Based Resource System-Level Dynamics Intern

Advisors: Dr. Soham Chakraborty and Dr. Jing Wang

May 2025 – Apr. 2026

Golden, CO

- Derived differential-equation models for various grid-following inverter control schemes; analyzed stability differences using small-signal analysis.
- Created a framework in MATLAB to automate parameter tuning of inverters to damp system-wide oscillations.
- Converted and validated a phasor-domain transmission system model into an EMT-domain digital twin for an island utility; applied root-cause analysis for sources of instability under varying weather conditions.

Colorado School of Mines

Graduate Research Assistant

Advisor: Dr. Qiuhua Huang

Jan. 2024 – May 2025

Golden, CO

- Evaluated methods for adjusting DER, IBR, and distribution system modeling detail for transmission and distribution dynamic co-simulation to reduce computational complexity.
- Implemented and scripted dynamic models, including DER_A, PVD1, induction motor, and AC motor models in InterPSS.

Colorado School of Mines

Mines Undergraduate Research Fellowship

Advisors: Dr. Qiuhua Huang and Asmaa Romia

Aug. 2023 – Jan. 2024

Golden, CO

- Developed an LSTM-based forecasting model for EV charging station demand and presented final results at the Mines Undergraduate Research Fellowship Symposium.

TEACHING AND MENTORING EXPERIENCE

Texas A&M University

Undergraduate Research Mentor

Jan 2026 – Present

College Station, TX

- Mentor an undergraduate freshman conducting power systems research in state-space modeling, transient stability, and small-signal methods.

Colorado School of Mines

Instructor, Introduction to Computer Science

Aug. 2024 – May 2025

Golden, CO

- Collaborated with Computer Science faculty to develop and deliver Python programming lectures three times per week to a class of approximately 75 students.
- Held review sessions, recorded supplementary lectures, and provided weekly office hours to support student

learning and engagement.

Colorado School of Mines

Jan. 2023 – May 2025

Recitation Leader, Teaching Assistant, Lab Assistant, Grader

Golden, CO

Courses: Information Systems; Embedded Systems; Electrical Devices and Circuits; Digital Design

- Delivered recitation lectures, held office hours, introduced laboratory activities, guided student lab projects, and graded student lab/homework assignments.

CONFERENCE AND POSTER PRESENTATIONS

Transmission and Distribution Dynamic Co-Simulation Contingency Analysis with Adaptive Distribution System Modeling Dec. 2024

O. Cornmesser, Q. Huang. *PSERC Dec. 2024 Industry and Advisors Meeting.*

Identifying Regions of Detailed DER Modeling for Transmission System Dynamic Contingency Analysis Using Short-Circuit Analysis Jul. 2024

O. Cornmesser, Q. Huang. *IEEE Power and Energy Society General Meeting 2024.*

Adaptive Transmission and Distribution Modeling and Simulation Tools for Supporting Integration and Operation of High Penetration of DER May 2024

O. Cornmesser, Q. Huang, A. Shankar Kar, V. Ajjarapu, Z. Zhou, M. Wu. *PSERC May 2024 Industry and Advisors Meeting.*

Advanced LSTM Forecasting for EV Charging Station Demand

Apr. 2024

O. Cornmesser, A. Romia, Q. Huang. *Mines Undergraduate Research Fellowship Symposium.*

INDUSTRY EXPERIENCE

POWER Engineers, Inc.

May 2023 – Dec. 2023

Electrical Engineering Intern

Lakewood, CO

- Developed substation designs through trip and close schemes, one-line and three-line schematics, wiring diagrams, SCADA communications, grounding grid plans, and related deliverables.

SERVICE AND LEADERSHIP

IEEE Texas Power and Energy Conference (TPEC)

Feb. 2026

Session Chair, Paper Reviewer, and Volunteer

- Reviewed conference paper submissions in power systems, providing technical feedback to support the peer-review process.
- Served as session chair and supported a student-run power systems conference connecting students, faculty, researchers, and industry professionals.
- Assisted with website development, logistical setup, and session operations.

Colorado School of Mines Varsity Golf Team, Captain

Aug. 2021 – May 2024

- Led teammates through athletic and academic responsibilities while representing Colorado School of Mines in collegiate competition.

AWARDS, HONORS, AND FELLOWSHIPS

- Chancellor's National Academy STEM Ph.D. Fellowship Program, Fall 2025 – May 2029
- Mines Undergraduate Research Fellow, Fall 2023 and Spring 2024.
- Summa Cum Laude, Colorado School of Mines, 2024.
- Dean's List, Colorado School of Mines.

PROFESSIONAL MEMBERSHIPS

- IEEE Power & Energy Society, Student Member, Jan. 2024 – Present.
- CIGRE, Student Member, Jan. 2024 – Present.

TECHNICAL SKILLS

Power Systems Software	PSSE, PSCAD, PowerWorld, OpenDSS, Simulink, ETAP, InterPSS
Programming and Data Analysis	Python, MATLAB, C/C++, Java, JavaScript, Git, LaTeX
Machine Learning Tools	PyTorch, TensorFlow, scikit-learn, Pandas, NumPy